

Curriculum Format

Programme Name: B. Tech. Computer Science and Engineering
Programme School: Computer Science and Engineering (SCOPE)
Curriculum applicable to (for admitted year
xxxx-xx format): AY 2023-24
Curriculum Version: 2.2
Approved by & Date 12th Academic Council, 25-05-2024

University Core Min. Credit Requirements ----->

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Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
	Projects and Internships	All are compulsory courses except for choices				24			
BIC4002		Industrial Internship/ Senior Design Project	2	0	0	12	None	None	Engineering
CAP4001		Capstone	1	0	0	6	None	None	Engineering
SIT1001		Summer Internship	1	0	0	2	None	None	Engineering
ECS2002		Engineering Clinics - System Design	1	0	4	2	None	None	Engineering
ECS3001		Engineering Clinics - Real Time System (or)	1	0	4	2	ECS1002	None	Engineering
MEC1002		Engineering Graphics	1	0	4	2	None	None	Engineering
	Engineering Foundation	All are compulsory				19			
CSE1012		Problem Solving using Python	2.0	3	2	4	None	SWE1004	Engineering
CSE1022		Introduction to Programming	1.0	2	2	3	None	None	Engineering
CSE2005		Object Oriented Programming using JAVA	4.1	3	2	4	CSE1022	CSE2015, SWE2005	Engineering
CSE2001		Data Structures and Algorithms	1.2	3	2	4	None	None	Engineering
ECE1002		Fundamentals of Electrical and Electronics Engineering	1.1	3	2	4	None	None	Engineering
	Humanities and Management	All are compulsory courses except for choices				24			
MGT1040		Entrepreneurship	1	2	0	2	None	None	Management
MGT1001		Ethics and Values	2	2	0	2	None	None	Humanities
STS1002/ STS1004		Fundamentals of Aptitude (or) Introduction to Problem Solving	1.1	3	0	3	None	None	Humanities
STS1007/ STS1009		Arithmetic problem solving (or) Introduction to quantitative, logical and verbal ability	1	3	0	3	STS1002/STS1004	None	Humanities
STS2006/ STS2008		Getting started to skill enhancement (or) Numerical ability and cognitive intelligence	1.1	3	0	3	STS1007/STS1009	None	Humanities
STS2007/ STS2009		Enhancing problem solving skills (or) Advanced aptitude and reasoning skills	1	3	0	3	STS2006/STS2008	None	Humanities

Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
		Any two English courses depending on SKEPT result				6			
ENG1001		English for Essential Communication	1.1	2	2	3	EPT (Not Cleared)	ENG1005	Humanities
ENG1002		English for Effective Communication	1.1	2	2	3	EPT (Cleared) /ENG1005	ENG1006	Humanities
ENG2001		English for Professional Communication	1.1	2	2	3	ENG1002	ENG2004	Humanities
		Any one Foreign Language course				2			
FRL1001		Basic French	1	2	0	2	None	None	Humanities
FRL1004		Basic Spanish	1	2	0	2	None	None	Humanities
FRL1005		Basic German	1	2	0	2	None	None	Humanities
FRL1006		Basic Japanese	1	2	0	2	None	None	Humanities
	Science Basket	All are Compulsory				20			
MAT1001		Calculus for Engineers	1	3	2	4	None	MAT1008	Science
MAT1002		Applications of Differential and Difference Equations	2	3	2	4	MAT1001	None	Science
MAT1011		Applied Statistics	2	3	2	4	None	MAT1014	Science
PHY1008		Modern Physics	2	3	2	4	None	None	Science
CHY1009		Chemistry and Environmental Studies	1	3	2	4	None	None	Science
		Co/Extra Curricular [Compulsory]	2			2	None		
Programme Core Min. Credit Requirements ----->							40		
Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
ECE1003		Digital Logic Design	1.1	3	2	4	ECE1001/ECE1002	None	Engineering
MAT1003		Discrete Mathematical Structures	2.0	4	0	4	None	MAT1007	Science
CSE3004		Design and Analysis of Algorithms	1.2	3	2	4	CSE2001	CSE3023, SWE2009	Engineering
CSE2008		Operating Systems	1.2	3	2	4	None	SWE2007, CSE2017	Engineering
CSE1005		Software Engineering	1.0	3	2	4	None	SWE1002	Engineering
CSE4004		Web Technologies	4.0	3	2	4	None	SWE4001	Engineering
CSE2007		Database Management Systems	2.1	3	2	4	CSE2001	SWE2006	Engineering
CSE3003		Computer Networks	2.0	3	2	4	CSE2005	SWE3001, CSE3025, ECE3003	Engineering
ECE2002		Computer Architecture and Organization	2.0	4	0	4	ECE1003	None	Engineering
CSE1008		Theory of Computation	2.0	4	0	4	None	CSE1013	Engineering

Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
Specialization Electives Min. Credit Requirements ----->						22			
Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
		Any course to fulfill the basket requirements							
CSE1006		Foundations for Data Analytics	1.1	2	2	3	None	None	Engineering
CSE1007		Introduction to Cryptography	2.0	3	2	4	None	None	Engineering
CSE2003		Requirements Engineering Management	1.0	3	0	3	CSE1005/SWE1002	SWE2003	Engineering
CSE2009		Soft Computing	1.1	3	2	4	None	None	Engineering
CSE2010		Secure Coding	1.0	3	2	4	CSE2001	None	Engineering
CSE2011		Cyber Security and Digital Forensics	1.0	3	2	4	CSE1007/SWE3003	None	Engineering
CSE2012		Knowledge Representation and Reasoning	1.0	3	2	4	None	None	Engineering
CSE2013		Information Theory and Coding	2.0	4	0	4	MAT1011	None	Engineering
CSE3001		Agile Development Process	1.1	3	2	4	CSE1005/SWE1002	None	Engineering
CSE3002		Artificial Intelligence	2.1	3	0	3	CSE1012/SWE1004	None	Engineering
CSE3005		Distributed Systems	1.0	3	2	4	CSE2008	None	Engineering
CSE3006		Data Visualization	1.0	3	2	4	CSE1006	None	Engineering
CSE3007		Information Retrieval	1.1	3	2	4	CSE2001	None	Engineering
CSE3008		Introduction to Machine Learning	3.0	3	2	4	None	CSE2019	Engineering
CSE3009		NoSQL Database	2.0	3	2	4	CSE2007/SWE2006	None	Engineering
CSE3010		Network Design and Performance Evaluation	1.0	3	2	4	CSE3003	None	Engineering
CSE3011		Network Programming	1.0	3	2	4	CSE3003	None	Engineering
CSE3012		Network Security	4.0	3	2	4	CSE3003/ ECE2006	None	Engineering
CSE3013		Secure Group Communications	1.1	3	0	3	CSE1007/SWE3003	None	Engineering
CSE3015		Natural Language Processing	2.1	3	2	4	CSE3008	None	Engineering
CSE3018		Software Configuration Management	1.1	3	0	3	CSE1005/SWE1002	None	Engineering
CSE3019		Software Quality and Reliability	1.0	3	0	3	CSE1005/SWE1002	None	Engineering
CSE3020		Product Definition and Validation	1.0	3	0	3	CSE1005/SWE1002	None	Engineering
CSE3033		Software Verification and Validation	1.0	3	2	4	CSE1005/SWE1002	SWE4002	Engineering
CSE4001		Cloud Computing	1.1	3	0	3	CSE3003	SWE4004	Engineering
CSE4003		Parallel Computing	2.1	3	2	4	CSE2008/ECE2002	None	Engineering

Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
CSE4005		Data Warehousing and Data Mining	1.1	3	2	4	CSE1006	None	Engineering
CSE4006		Deep Learning	4.1	3	2	4	None	None	Engineering
CSE4007		Digital Image Processing	1.0	3	2	4	None	None	Engineering
CSE4008		Social Network Analysis	1.0	3	2	4	CSE1006	None	Engineering
CSE4009		Wireless and Mobile Computing	1.0	3	0	3	CSE3003/SWE3001	None	Engineering
CSE4010		Wireless and Mobile Security	1.0	3	0	3	CSE3012	None	Engineering
CSE4011		Internet of Things	1.0	3	2	4	None	None	Engineering
CSE4012		UI/UX User Experience Design	1.0	3	2	4	None	None	Engineering
CSE4013		Reverse Engineering	1.0	3	0	3	CSE1005/SWE1002	None	Engineering
CSE4014		Team Software Process	1.0	3	0	3	CSE1005/SWE1002	None	Engineering
CSE4015		Network and System Administration	2.0	3	2	4	SWE3001/CSE3003	None	Engineering
CSE4016		DataBase Administration	1.1	3	2	4	CSE2007/SWE2006	None	Engineering
CSE4017		Healthcare Analytics	1.1	3	0	3	CSE1006	None	Engineering
CSE4018		Computer Vision and Robotics	1.1	3	2	4	CSE4007	CSE4047	Engineering
CSE4019		Applications of Artificial Intelligence	1.0	3	0	3	CSE3002	None	Engineering
CSE4020		Agent Based Intelligent Systems	1.1	3	2	4	CSE3002	None	Engineering
CSE4021		Introduction to Cognitive Modelling	1.1	3	2	4	CSE3002	None	Engineering
CSE4022		Modelling and Simulation	1.0	3	2	4	None	None	Engineering
CSE4031		Web Application Security	1.0	3	2	4	CSE3012	None	Engineering
CSE4037		Reinforcement Learning	2.0	3	2	4	CSE3008	None	Engineering
CSE4040		Mining Massive Data Sets	1.0	3	2	4	CSE4005	None	Engineering
CSE4043		Intrusion Detection and Prevention System	1.1	3	2	4	CSE3003	None	Engineering
CSE4046		Design Patterns	1.1	3	2	4	CSE2005/SWE2005	None	Engineering
CSE4047		Computer Vision	1.1	3	2	4	CSE4007	CSE4018	Engineering
CSE2025		AWS Solution Architecture	1.0	3	0	3	None	None	Engineering
CSE2027		Software Design and Architecture	1.0	3	2	4	CSE1005/SWE1002	SWE2004, CSE3014	Engineering
CSE2026		Object Oriented Analysis and Design	1.0	3	2	4	CSE1005/SWE1002	SWE2008, CSE2005	Engineering
CSE4042		Software Project Management	1.0	3	2	4	CSE1005/SWE1002	SWE3002	Engineering
CSE1021		Foundations of Blockchain Technology	1.0	3	0	3	None	None	Engineering
CSE2024		Blockchain Architecture Design	1.0	3	2	4	CSE1021	None	Engineering

Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
CSE3036		Cryptocurrency Technologies and Bitcoin	1.0	3	2	4	CSE1021	None	Engineering
CSE3035		Blockchain and Distributed Ledger Technology	1.0	3	2	4	CSE1021	None	Engineering
CSE3037		Public Key Infrastructure & Trust Management	1.0	3	2	4	CSE1007/SWE3003	None	Engineering
CSE3038		Smart Contracts	1.0	3	2	4	CSE1021	None	Engineering
CSE3040		Incident Response Management System	1.0	3	2	4	CSE2011	None	Engineering
CSE3041		Malware Analysis	1.0	3	2	4	None	None	Engineering
CSE4043		Vulnerability Assessment and Penetration Testing	1.0	3	2	4	CSE2011	None	Engineering
ECE2014		Microprocessors and Microcontrollers	1.0	3	2	4	ECE1003	None	Engineering
ECE4003		Embedded Programming	1.0	3	2	4	ECE2004	None	Engineering
MAT2003		Optimization Techniques	1.0	3	2	4	MAT1001/MAT2004	None	Science
University Electives Min. Credit Requirements ----->						9			
Course Code	Category	Course Title	Ver.	T	P	C	Pre/Co-Requisite	Anti-requisite	Course Discipline
Any Course after fulfilling Programme Core & University Core requirements and without dulplicity can be taken as University Elective									

CSE Core	Credits
Engineering Foundation	19
Science Basket	20
Projects and Internships	24
Humanities & Management + Clubs	26
Total UC	89
Program Core	40
Program Electives	22
University Elective	9
Total Credits	160